project requirements

✅ Admin Management Module – Requirements

1. Functional Requirements

* Admin can add new admin users
* Admin can view all patients and doctors
* Admin can activate or deactivate users
* Admin can assign and update user roles
* Admin can delete user accounts if required
* Admin can view all appointments in the system
* Admin can identify and resolve scheduling conflicts
* Admin can cancel or reschedule appointments
* Admin can update doctor profiles and availability
* Admin can monitor overall system performance

2. Controller Requirements (AdminController)

* addAdmin()
* manageUsers()
* viewSystemLogs()
* viewAllAppointments()
* resolveAppointmentConflicts()

3. Service Requirements (AdminService)

* Validate admin authentication and authorization

* Handle role-based access control
* Manage user lifecycle operations
* Fetch and process system logs
* Coordinate with Appointment Service for conflict handling

4. Data Requirements (Admin Entity)

* adminId (Primary Key)
* name
* email
* role

5. Security Requirements

* Only authorized admins can access admin functions
* Role-based access must be enforced
* All admin operations must be logged
* Secure session handling and timeout

6. Non-Functional Requirements

* Fast response time for admin operations
* Support multiple admins simultaneously
* Ensure system reliability and availability
* Maintain data integrity and audit trails

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✅ Notification Module – Requirements

1. Functional Requirements

* System sends appointment confirmation notifications
* System sends appointment cancellation notifications
* System sends appointment reminder notifications
* Notifications are sent automatically based on appointment status
* Notifications can be sent via email
* SMS notification support (optional)
* System avoids duplicate notifications
* Notifications are logged for auditing

2. Controller Requirements (NotificationController)

* sendConfirmationNotification()
* sendAppointmentReminder()
* sendCancellationNotification()

3. Service Requirements (NotificationService)

* Generate notification messages dynamically
* Identify recipients (patient or doctor)
* Integrate with email/SMS services
* Retry notification delivery on failure
* Maintain notification delivery logs

4. Business Rule Requirements

* Notifications are sent only for valid appointments
* Reminder notifications are not sent for canceled appointments
* Notifications must follow configured schedules
* Delivery status must be stored in the database

5. Data Requirements (Notification Entity)

* notificationId (Primary Key)
* recipientId
* message
* notificationType (CONFIRMATION, REMINDER)
* sentDate

6. Non-Functional Requirements

* Timely notification delivery
* Support high volume of notifications
* Secure handling of personal contact details
* Reliable and fault-tolerant notification mechanism